

Q1)

A)

1. b
2. a
3. c
4. b

pc : is there an object

bx, by, bh, bw : bounding box

c1, c2, c3: predicted class

5. b

$$IoU = \frac{Overlap}{Union} = \frac{1 \times 1}{(2 \times 2) + (2 \times 3) - (1 \times 1)} = \frac{1}{4 + 6 - 1} = \frac{1}{9}$$

6. d
7. d
8. b (مش علينا؟)
9. d
10. مش شايف حاجة.
11. a

$$YOLO\ output = Grid \times Grid \times (5 \times B + C)$$

$$Grid = 19 \quad C = 20 \quad B = 5$$

$$Output = 19 \times 19 \times (5 \times 5 + 20) = 19 \times 19 \times 45$$

12. c
13. c
14. c
15. مش علينا.
16. d
17. d
18. d

$$N > \frac{\log(1-p)}{\log(1-(1-e)^s)}$$

$$e = 0.3 \quad p = 0.8 \quad s = 1 \text{ (translational model needs min 1 samples)}$$

$$N > \frac{\log(1-0.8)}{\log(1-(1-0.3)^1)} = 1.33 \approx 2$$

19. مش علينا.
20. a
21. d
22. مش علينا.

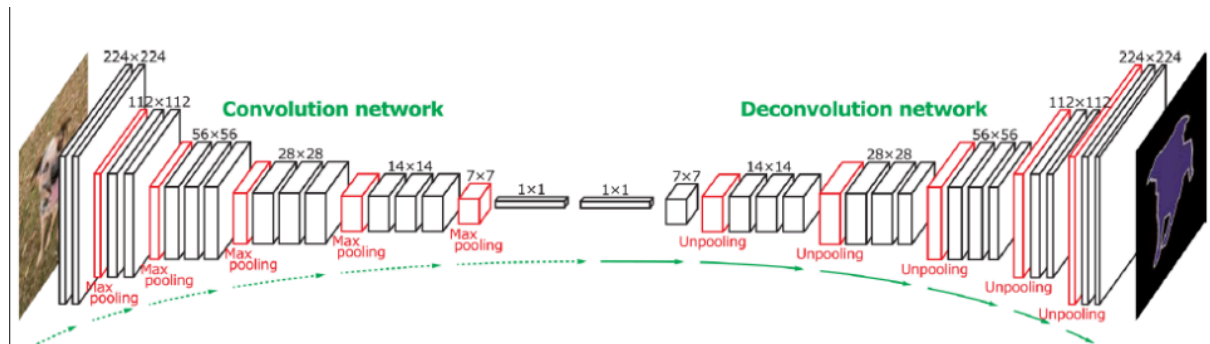
Q2)

A) مش علينا

B)

i. Semantic Image Segmentation

ii. Deconvolution Network for Semantic Segmentation



Input: an image

Output: A Map containing the class of each pixel of the input image

iii. Instance Image Segmentation

iv. The target image should be labelled such that the objects each have a bounding box

Q3)

I.

	Method of Region Proposal	Number of ConvNets	Object Classification Function	Speed
R-CNN	Selective search	2000	SVM	Slow
Fast R-CNN	Selective Search	1	Softmax	Fast
Faster R-CNN	Region proposal network	1	Softmax	Faster
YOLO	-	1	-	Realtime (45 fps)

II. مش علينا

Q4) مش علينا